

Alexander Müller

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PROFILE

MSc. Artificial Intelligence student at the University of Groningen with research interests in AI safety, such as mechanistic interpretability and multi-agent systems. Co-directing the AI Safety Initiative Groningen and publishing work on LLM compliance benchmarking, activation steering, and multi-agent work. Experienced educator ranked 3rd out of ~50 teaching assistants in 2025.

EDUCATION

- University of Groningen** Groningen, The Netherlands
Bachelor of Science – Artificial Intelligence September 2022 – July 2025
 - Relevant Courses:** Algorithms and Data Structures, Object-Oriented Programming, Linear Algebra, Statistics, (Multivariable) Calculus, Neural Networks, Reinforcement Learning, Uncertainty in ML, Analysis, Advanced Logic, Ethics in AI, Cognitive Psychology, Neuroscience
 - GPA:** 8.9/10
 - Thesis:** [Machine Learning for Peptide Property Prediction: A Robust Nested Cross-Validation Pipeline with Mechanistic Interpretability](#). Conducted the first case study of mechanistic interpretability in the domain of chemistry. Grade: 9/10.
- Master of Science – Artificial Intelligence* September 2025 – July 2027
 - Relevant Courses:** Advanced Machine Learning, Deep Learning, Proof Theory and Modal Logic, Logical Aspects of Multi-Agent Systems, Responsible AI, Design of Multi-Agent Systems, Computational Game Theory, Pattern Recognition, Methodology in AI
 - GPA:** 8.7/10

PUBLICATIONS

- I. Lichkovski, **A. Müller**, M. Ibrahim, T. Mhundwa. “EU-Agent-Bench: Measuring Illegal Behavior of LLM Agents Under EU Law.” *arXiv preprint*, 2025.
- A. Müller**, J. Cardenas-Cartagena, R. Pollice. “Uncovering Internal Prediction Mechanisms of Transformer-Based Chemical Foundation Models.” *ChemRxiv*, 2025.
- S. Abreu, J. Postmus, **A. Müller**, J.L. Ferrao, I. Lichkovski, K.F. Michalak, et al. “From Steering Vectors to Conceptors: Compositional Affine Activation Steering for LLMs.” 2025.
- A. Müller**, A. Golicins, G. Lesnic. “Collective Deliberation for Safer CBRN Decisions: A Multi-Agent LLM Debate Pipeline.” *AISIG & Apart Research*, 2025. [Code]

WORKING EXPERIENCE

- Co-Director, AI Safety Initiative Groningen (AISIG):** Aug. 2025 – Ongoing
 - Leadership:** Co-directing a student-led AI safety organization of [X] members. Organized [X] workshops, seminars, and reading groups reaching [X]+ participants.
 - Research:** Supervising [X] interdisciplinary research projects on LLM compliance, activation steering, and multi-agent safety. Collaborated with Apart Research on a hackathon project.
 - Outreach:** Represented AISIG at EAGxAmsterdam and aiGrunn conferences. Delivered invited talk for Turn.io staff on LLM chatbot risks and mitigations.
- Founder & Lead, NeurAlignment Research & Discussion Group:** Nov. 2024 – Ongoing
 - Research Coordination:** Founded and leading a cross-level group (BSc–PhD) exploring how neuroscience can inform AI alignment. Currently supervising three research projects on [topics].
- Teaching Assistant:** Sep. 2023 – Feb. 2025
 - Courses:** Introduction to Artificial Intelligence (2×), Basic Scientific Skills (2×), Cognitive Psychology, Introduction to the Brain — led tutorials, labs, and assessments across six course instances.
 - Recognition:** Awarded **3rd place Teaching Assistant of the Year** (out of ~50 TAs) in 2025.
- Mentor for First-Year AI Students:** Sep. 2023 – Feb. 2024; Sep. 2024 – Feb. 2025
 - Student Support:** Mentored [X] first-year AI students across two cohorts, guiding them through the Dutch university system and student life.
- Private Tutor:** Mar. 2019 – Feb. 2024
 - Personalized Instruction:** Tutored [X]+ high school students in Mathematics, Chemistry, and Physics over five years, developing tailored lesson plans for major national exams.

SELECTED PROJECTS

- The Wisdom of the LLM Crowd:** Simulated deliberative groups of five LLMs on 1,000 Moral Machine dilemmas to test whether collective deliberation improves moral alignment. Found that debate-induced extremization collapsed nuanced priors into simplistic heuristics. *Design of Multi-Agent Systems, UG*.
- Intervention Analysis on the Latent Space of VAEs:** Investigated how training data diversity influences uncertainty and reconstruction in Variational Autoencoders. Applied intervention analysis on MNIST/FashionMNIST to identify which latent dimensions drive reconstruction error. *Advanced Machine Learning, UG*.
- Strategic Voting via Dynamic Epistemic Logic:** Modeled strategic voting in repeated elections using S5 epistemic logic with polls as public announcements. Showed how a strategic voter can manipulate outcomes while inadvertently creating common knowledge. *LAMAS, UG*.
- Sequent Calculus Proofs for Modal Logic:** Derivations for propositional and modal logic covering standard axioms, cut elimination, and S4/S5 systems. *Proof Theory and Modal Logic, UG*.

SKILLS

- **Languages:** Dutch (native), English (C2 — Cambridge certificate), German (B2)
- **Programming:** Python, C, Java, R
- **ML & Data Science:** PyTorch, Scikit-learn, Keras, Pandas, NumPy, Matplotlib, Gymnasium
- **Tools:** Git/GitHub, $\LaTeX$, Jupyter, Linux
- **Web:** Astro, Tailwind CSS, HTML/CSS
- **Writing:** Academic writing (C2 Cambridge certificate) and informal writing ([Substack](#))

TALKS

- [How to Run an AI Safety Initiative? \(Dec. 2025\)](#): One-hour workshop for 30 attendees at EAGxAmsterdam on founding and scaling a university AI safety initiative.
- [AI Safety at aiGrunn Conference \(Nov. 2025\)](#): 30-minute talk for 50 software developers on risks from advanced AI, focusing on superintelligence.
- [AI Safety at Turn.io \(Aug. 2025\)](#): One-hour talk and Q&A for 30+ staff on risks from LLM-powered chatbots, with actionable mitigation proposals.